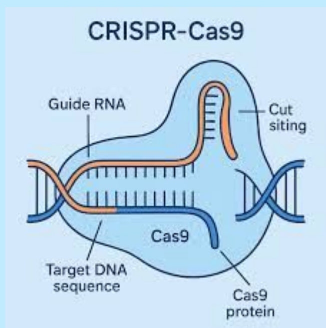
# Pre-Research Question:
# Can CRISPR gene editing reduce sickle cell disease symptoms while maintaining safe off-target risk?

**Figure 1**

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## Intro & Background Info:

<u>The potential benefits of using gene editing procedures to treat fetal sickle cell disease outweigh the risks as they include prenatal opportunities to prevent irreversible damage, reduce the chances of future gene therapies' ineffectiveness, and curb the possibility of immune system rejection.</u>

As the name suggests, gene editing involves making changes to the DNA. Initially, scientists used it to change plants, bacteria, and animals. But now, gene editing can also be used to cure diseases. According to the NIH, "In 2015, scientists successfully used somatic gene therapy when a one-year-old in the United Kingdom named Layla received a gene editing treatment to help her fight leukemia, a type of cancer" (Genome.gov, 2019). In recent years, another disease being targeted is sickle cell disease, which you probably heard about in a biology class. It is a condition that entails sickle-shaped red blood cells; in sufferers, blood cannot flow smoothly. This can lead to pain, organ damage, and other serious health issue (National Heart, Lung, and Blood Institute, 2024). We chose to research this topic because sickle cell disease treatment is an early example of the capabilities of gene editing, and its possibilities are rapidly expanding. After the first successful CRISPR-based treatment of sickle cell disease in 2019, the patient was cured (Stein 2021). This achievement indicates a promising future for the field of gene editing and hints at the benefits it could provide to similar patients moving forward. The promise shown by this new field is what drove us to delve deeper into research, hoping to learn more about both the potential benefits and the limitations of this new form of treatment.

## Methods/Definitions/Context:

Using the NC Live database, we kept our searches recent (2019 onward) and used terms like "gene editing", "fetal", and "sickle cell disease" to narrow down results. We included both articles that discussed the emerging field of fetal gene editing in general (its drawbacks, health/ethical concerns, and potential benefits in terms of immunotolerance and disease prevention) as well as sources that pertained specifically to sickle cell disease and its treatment in utero.

Vital contextual vocabulary:

- In vivo versus ex vivo gene editing: *In vivo* involves gene therapies inserted directly into the fetus with the aid of a viral-based system. *Ex vivo* editing involves the removal of cells and the administration of gene therapy outside of the fetus, followed by the subsequent reinsertion of the edited cells into the fetus (Sagar & David, 2024).
- Monogenic: In this context, a disease that only affects a single gene (e.g. sickle cell disease). These diseases are easier to treat with precise gene editing (Shanahan et al., 2021).
- CRISPR-Cas9: A gene editing tool that can be used to specifically target a certain section in the genome (Shanahan et al., 2021).

## Results & Data Visualization, some Analysis:

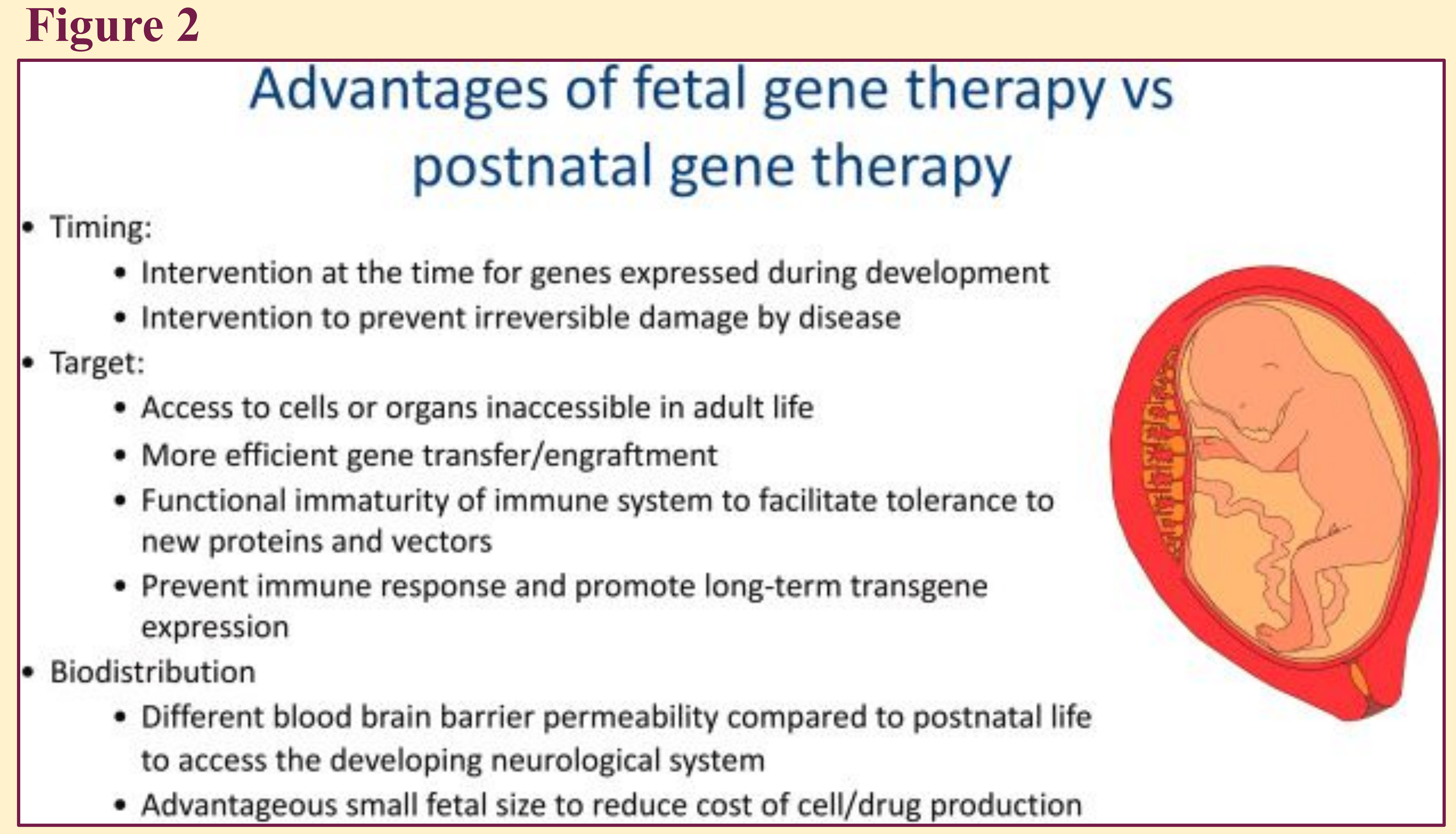
☆ "Prenatal gene therapy can treat monogenic disorders even before the pathology of the disease begins, thereby significantly decreasing morbidity and mortality."(Peddi, 2025)

☆ "Kiran Musunuru at the University of Pennsylvania and his colleagues have tested gene editing in monkeys and found that it was far more effective when done before birth. 'What we found was astounding,' he says. 'It opens up the opportunity to treat diseases that have been very hard to treat after birth.'"(Le Page, 2024)

☆ "Compared with postbirth gene therapy, fetal gene therapy offers several opportunities to circumvent implementation barriers… The fetal period offers an opportunity to capitalize on the developmental window when immune tolerance is greater than it is postnatally." (Shanahan et al., 2021)

☆ One barrier that gene therapy on the utero bypasses is intolerance and rejection.
- "Immunoglobulin (Ig) G isotypes can render subsequent gene therapy episodes ineffective. However, during the fetal period, IgG isotype class switching generally does not occur, and the fetus is limited largely to IgM production. Fetal gene therapy thus provides a marked advantage over the only current curative option, which carries the significant risk of graft-vs-host disease."(Shanahan et al., 2021)

**Figure 2**




**(https://www-sciencedirect-com.proxy122.nclive.org/science/article/pii/S1521693424000968#abs0010, 2024)**

❖ After reading this information, it can be understood that the process of editing the complexion of genes prenatally compared to postnatally brings about a higher rate of success due to being easier and more efficient.

❖ This brings a necessity to addressing diseases such as sickle cell early on.

❖ While technology and knowledge in this field is still developing and improving, many scientists are further proving that the benefits of operating on the fetus are enhancing and the risks continue to reduce as time passes.

## Analysis & Conclusion:

While there are certainly risks associated with the rapidly-emerging procedure of gene editing to treat sickle cell disease, its potential to increase the health and quality of life of patients is also notable. Sickle cell disease causes lifelong damage and significant pain in sufferers, and its prognosis involves a decreased life expectancy (Shanahan et al., 2021). Preventative treatments like fetal gene editing can be a valuable tool in reducing the harm these patients will endure throughout their lifetimes.

Attempting either in vivo or ex vivo gene editing does pose certain risks (such as being off-target and causing unintended genetic consequences), but when performed correctly, these procedures can take advantage of unique circumstances in the window of fetal development (Sagar & David, 2024). Evidence from clinical studies indicates that fetuses may generate less immune resistance to the intrusion of a viral carrier agent and produce fewer IgG isotypes that would cause future procedures to be useless (Shanahan et al., 2021). In addition, research shows that early intervention may prevent damage that the disease would ordinarily inflict before birth (Le Page, 2024). Combined, these factors may result in a less dangerous and more effective gene editing procedure than those performed on postnatal subjects, and lessened risk as compared to the dangers faced by patients with sickle cell disease left insufficiently treated.

Certain limitations, like the difficulty of identifying the disease quickly enough to take advantage of the fetal period of heightened immunotolerance, the potential of harming the fetus or parent, and the cost of the procedure, do impact its opportunities of widespread implementation (Shanahan et al., 2021). However, as technologies advance in the years to come and procedures become more widely available and reliable at a reasonable cost, the practice of fetal gene editing to offer prevention for sickle cell disease may become increasingly common and could improve the lives of patients around the world.

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